

Biosorption of Methylene Blue from an aqueous solution using French green beans and Lotus Stem

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Abstract-This paper explores the potential of French green beans (*Phaseolus Vulgaris*) and Lotus Stem (*Nelumbo Nucifera*) as potential low-cost adsorbents for the removal of Methylene Blue from wastewater. The studies were conducted with variation in parameters of pH, biomass concentration, contact time and initial dye concentration. The nature of the possible adsorbent and metal ion interactions was examined by the FTIR technique. Adsorption of Methylene Blue onto French green beans could be described by the Freundlich model while Langmuir model with a maximum adsorption capacity of 62.5 mg/g was followed by Lotus Stem. The experimental data was modelled using pseudo-first and pseudo-second-order kinetic equations. Adsorption kinetics could be described by pseudo-second order equation.

Keywords- Biosorption, French green beans, Isotherm, Kinetics, Lotus Stem, Methylene Blue

1. Introduction

Dyes are coloured organic compounds used to impart value to the final product during the manufacturing phase. While a part of it is used up, the rest of it is discharged into the municipal waste streams as effluent from industries such as Textile, Leather, Cosmetics, and Pharmaceuticals. Not only is the discharged effluent aesthetically unpleasing, it causes a number of environmental complications such as increase in toxicity, high chemical oxygen demand (COD), biological demand (BOD) of the effluent and reduction in light penetration[1]. Such effects are harmful for the biotic component in the water. Therefore considerable research has been carried out to find effective ways for the removal of dyestuff from wastewater.

Methylene Blue (MB) is a commonly dye which is widely used to make hair colours, cotton dyes and. Although it is not hazardous, it can still have some deleterious effects such as nausea, vomiting [2] and jaundice [3].

The prevailing methods of treating wastewater [4-6] have lower efficiency as compared to adsorption. Activated carbon is the most popular adsorbent. Even though its simplicity and insensitivity to toxicity provide it with an advantage compared to the conventional techniques, high exorbitant has prevented its unconditional use. Hence scientists are looking for viable adsorbents. Recently numerous investigations have been made on the use of agricultural products [7-12] for the removal of Methylene Blue from aqueous solution but none have reported the use of French Beans (FB) and Lotus Stem (LS). In this study batch investigations were performed to investigate the capability of French green beans and Lotus stem in removing MB from an aqueous solution.

2. Material & Methods

2.1 Biosorbent preparation & Characterization

French green beans and Lotus stem which were acquired from the local market were extensively washed with distilled water to remove impurities and coloration. The washed sorbents were then dried in an oven set at 105°C until they were dry. The dried sorbents was crushed and passed through a sieve of pore size 100 µm to get a uniform grain size distribution.

The surface functional groups of FB and LS were detected by Fourier transform infrared (FTIR)

spectroscopy FTIR-2000, PerkinElmer. The spectra were recorded from 4000 to 450 cm^{-1} .

2.2 Preparation of dye solution

The stock solution of Methylene Blue was prepared by dissolving predetermined amounts of MB in distilled water, to achieve a concentration of 1000 mg/L. Different initial concentrations of MB were prepared by diluting the stock solution. The pH was adjusted using 0.1 M H_2SO_4 and 0.1 M NaOH.

2.3 Batch adsorption experiments

All batch experiments are conducted with the adsorbent samples in a beaker on a jar apparatus at 200 rpm to interpret the optimum conditions of pH, adsorbent and MB concentration.

The effect of pH on the adsorption rate of MB onto FB and LS are determined by equilibrating the adsorption mixture with dried sorbent and 250 mL of 50 mg/L MB solution at different pH values between 3-9. The effect of the adsorbent dose is studied using 50 mg/L MB solution and adsorbent dosage range between 0-10 mg/L.

10-1000 mg/L concentration dye solutions are used for the estimation of initial dye concentration on adsorption. The optimum doses of FB and LS which are determined as 8g/L and 6 g/L respectively are used to run the adsorption experiment for the time interval 5-150 min under isothermic conditions of 25°C. The solution is passed through Whatman Filter paper No 1. after the procedure is finished to obtain the sorbed residue. The supernatant is examined to determine the residue MB ion concentration.

The experimental data is then modelled for adsorption isotherms as well as kinetic models.

2.4. Analytical methods

The final MB concentrations of the solutions were determined by using a UV-Vis spectrophotometer (Hach DR 500, India) at 665 nm.

The concentration of dye retained in the sorbent phase (q_e , mg/g) was calculated from the expression:

$$q_e = \frac{(C_0 - C_e) \cdot V}{m} \quad (1)$$

where C_0 and C_e are the initial and final (equilibrium) concentrations of the dye in the solution (mg/L), V is the solution volume (L) and m is the mass of the sorbent (g).

For the calculation of the Methylene Blue rate adsorption (R_e), we used the following expression:

$$R_e(\%) = \frac{(C_0 - C_r) \cdot 100}{C_0} \quad (2)$$

where C_r is the residual concentration of Methylene Blue in the solution (mg/L).

3. Result & Discussion

3.1 FTIR Analysis

The FTIR spectra of unloaded and MB loaded FB as well as unloaded and MB loaded LS were taken and compared to obtain information on the nature of possible adsorbent-dye interaction.

The FTIR spectrum of FB (figure not shown) displays a broad and strong peak at 3354 cm^{-1} which indicates free or bonded O-H groups, a peak at 2939 cm^{-1} indicates the stretching of C-H of the CH_2 group. The symmetric stretching vibration of ionic carboxylic groups ($-\text{COO}-$) appears at 1629 cm^{-1} . The vibration at 1249 cm^{-1} is assigned to O-C-H, C-C-H, and C-O-H. The band observed at 1054 cm^{-1} can be assigned to the stretching vibration of C-OH of the alcoholic groups and carboxylic acids.

The FTIR spectra of LS (figure not shown) displays a broad and strong peak at 3386 cm^{-1} which indicates a free or bonded O-H groups, 2931 cm^{-1} indicates the stretching of C-H of the CH_2 group. The peak at 1640 cm^{-1} indicates carbonyl of non substituted amide and water. The peaks at 1423 cm^{-1} is assigned to OH, C-H, C-N and 1156-1018 cm^{-1} to C-O, C-O-C.

3.2 Effect of Operating Parameters

3.2.1 Effect of pH on dye removal

pH of a dye solution is an one important parameter affecting the adsorption capacity of an adsorbent as seen in fig 1 . Plot of Biosorption efficiency vs pH indicates that no adsorption of MB is observed up till pH 4. Beyond pH 4, the removal efficiency

continues to rise till it reaches its peak at pH. Further increase in pH causes a decrease in % MB removal. The lower adsorption at acidic pH could have due to the presence of excess H^+ , which provides competition to the dye cations for the available binding sites. Rise in pH makes the surface of FB and LS negatively charged, causing enhanced electrostatic interactions with the MB cations. A decrease in removal efficiency above pH 7 could be explained due to the presence of an additional mode of adsorption (ion exchange or chelation for example).

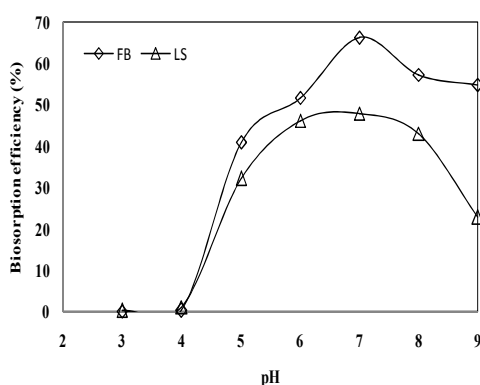


Figure 1. Effect of pH on adsorption of MB at 25°C

3.2.2 Effect of biomass concentration

Fig 2. illustrates the relationship between the biosorption efficiency and the biosorbent dose. The removal efficiency of MB increases rapidly for the given biosorbent doses, followed by a steady linear

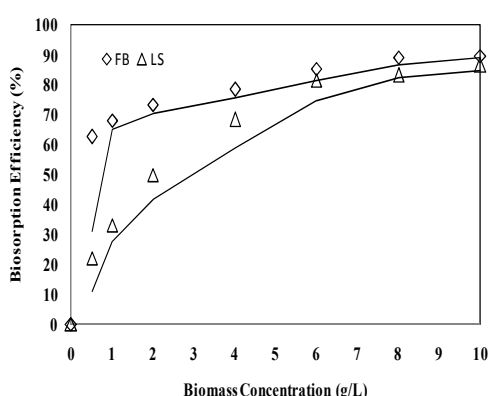


Figure 2. Effect of Biosorbent dosage on adsorption of MB at 25°C

relationship up till the optimum doses which are 8 g/L and 6 g/L for FB and LS respectively. Beyond these values of dose, the percentage increase in MB is not substantial to warrant a proportional increase in adsorbent dose. Increasing biosorbent dose causes an increase in surface area and the available

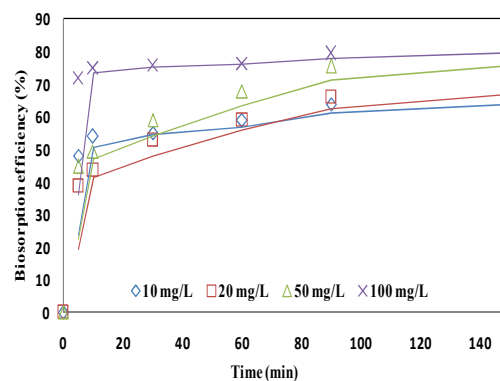


Figure 3. Effect of contact time on adsorption of MB by FB at 25°C

binding sites. Hence during the experimental process, 8 g/L and 6 g/L were chosen FB and LS as optimum doses for further analysis.

3.3.3 Effect of Contact time and Initial Concentration

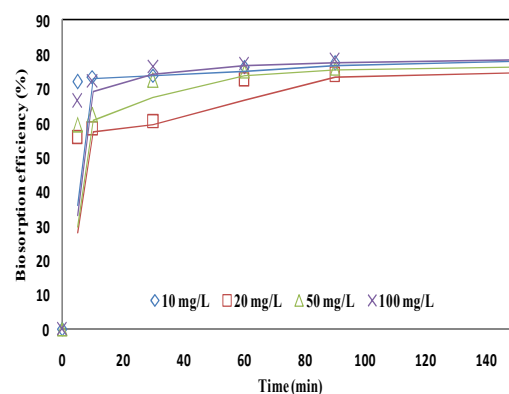


Figure 4. Effect of contact time on adsorption of MB by LS at 25°C

The adsorption capacity of MB removal by FB and LS vs time is shown in fig 3-4. In case of FB the amount of MB removed increases from 63.7 % to 79.4 % as the initial MB concentration increases while the amount of MB removed is independent of initial dye concentration in case of LS. The rate of percent dye removal is higher in the beginning due to a larger surface area of FB and LS being available for the adsorption of the MB. After the adsorbed material forms a 1 molecule thick layer, the capacity of the adsorbent is exhausted and subsequently the uptake rate is controlled by the rate at which the sorbate was transported from the exterior to the interior sites of the adsorbent particles. Based on the experimental data received, it is concluded that the equilibrium time is 90 minutes in the case of both FB and LS for initial concentration range of 10-100 mg/L.

3.3.4 Adsorption Isotherms

Sorption equilibrium can be explained by a number of models available in literature. In this work, three two-parameter models were used to characterize the isotherm data.

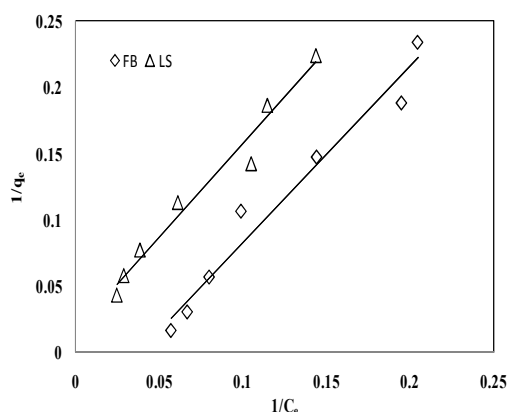


Figure 5. Langmuir isotherm of adsorption of MB at 25°C

The Langmuir isotherm model describes the monolayer sorption of the solutes onto the adsorbent surface. The adsorbent surface is assumed to have identical adsorption sites [13]:

$$\frac{1}{q_e} = \frac{1}{b \cdot q_{max}} \cdot \frac{1}{C_e} + \frac{1}{q_{max}} \quad (3)$$

where q is the metal uptake capacity (mg/g) and C_e is the concentration of metal ions in the solution (mg/L) at equilibrium, q_{max} (mg/g) and b (L/mg) are Langmuir isotherm constants.

Using the value of b from Langmuir equation, a dimensionless constant R_L is determined. The value of R_L is given by equation 4:

$$R_L = \frac{1}{1 + b \cdot C_0} \quad (4)$$

The values of R_L indicate the type of the Langmuir

isotherm to be irreversible ($R_L = 0$), favourable ($0 < R_L < 1$), linear ($R_L = 1$) or unfavourable ($R_L > 1$) [14].

The Freundlich isotherm is an empirical expression

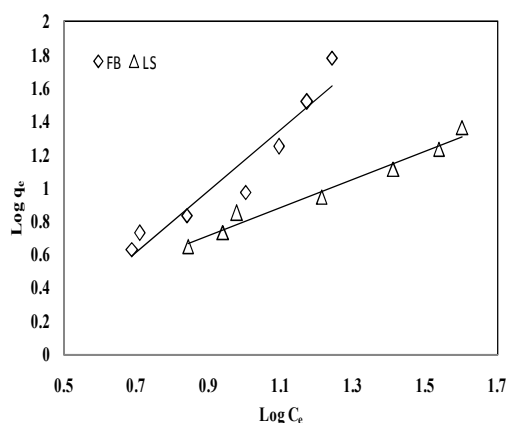


Figure 6. Freundlich/Halsey isotherm of adsorption of MB at 25°C

that takes into account the heterogeneity of the surface and is not restricted to the formation of a monolayer. The Freundlich model is expressed as follows [15]:

$$\text{Log } q_e = \text{Log } K_f + \frac{1}{n} \cdot \text{Log } C_e \quad (5)$$

where K is the biosorption equilibrium constant and n is a constant indicative of biosorption intensity. The value of slope $1/n$ lying between 0 and 1 is a measure of the adsorption intensity or surface heterogeneity. It becomes more heterogeneous as the value approaches closer to 0 [16].

The Halsey isotherm can be used to evaluate multilayer adsorption system [17]. It can be expressed as:

$$\log(q_e) = \frac{1}{n_H} \cdot \log K_H - \frac{1}{n_H} \cdot \log C_e \quad (6)$$

Table 1. Isotherm model predictions for adsorption of MB onto FB and LS at 25°C

	French green beans (FB)	Lotus Stem (LS)
Langmuir	q_0 (mg/g)	25
	b (L/mg)	0.0114
	R^2	0.97
Freundlich	K_F (L/g)	0.889
	n	1.179
	R^2	0.975
Halsey	K_H	1.151
	n_H	-1.179
	R^2	0.975

K_H and n_H are the Halsey isotherm constant and exponent, respectively. The plots of the linear form of Langmuir, Freundlich/Halsey adsorption isotherms for FB and LS obtained are illustrated in fig. 5-6.

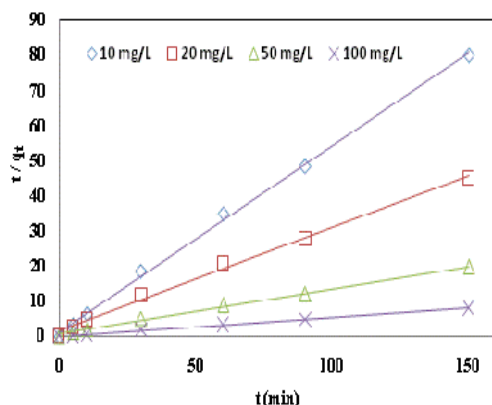


Figure 7. Pseudo Second Order kinetic model of adsorption MB by FB at 25°C

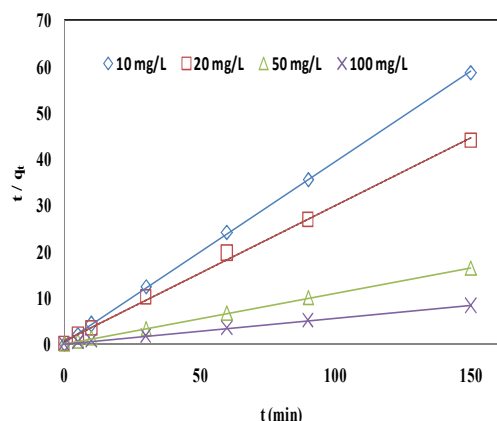


Figure 8. Pseudo Second Order kinetic model of adsorption MB by LS at 25°C

All of the isotherm model parameters for the adsorption of MB onto FB and LS are tabulated in table 1. In case of LS, based on values of R^2 , all the three isotherm models are able to describe the experimental data well. The maximum Langmuir adsorption capacity was found to be 62.5 mg/g.

The value of R_L calculated is 0.467 which means that the LS is favourable for adsorption of MB under the conditions studied. The constants K_F and n in the Freundlich model are 0.889 L/g and 1.179. Since the value of n is between 1 and 10, it suggests that the adsorption of MB ions is strong onto the LS surface. The Halsey parameters calculated K_H and n_H are 1.151 and -1.179

respectively. It is evident from the data that the surface of LS is made of homogeneous and heterogeneous adsorption patches. In the case of FB, although the plot of $1/C_e$ vs $1/q_e$ is a straight line, and based on values of R^2 , the values of the Langmuir constants are negative. This indicates the inadequacy of the isotherm model to explain the adsorption process, since these constants are indicative of the surface binding energy and monolayer coverage. The data is also fitted to the Freundlich model and has a value of $1/n > 1$. This may be indicative of cooperative adsorption [18]. The Halsey parameters calculated K_H and n_H are 2.34 and -0.541 respectively.

3.3.5. Adsorption Kinetics

The pseudo -first order kinetic model provided by Lagergren is given by the following equation [19]:

$$\log(q_e - q_t) = \log q_e - (K_1/2.303) \cdot t \quad (7)$$

This model describes the rate of sorption to be proportional to the number of sites unoccupied by the solutes. Here, q and q_e are the metal ion concentrations (mg /g) at any time (t) at the equilibrium (e), respectively, and K_1 is the first order rate constant (min^{-1}).

The pseudo-first order model is limited to regions of rapid biosorption. Hence its application over entire sorption range is not suitable. Ho's pseudo-second order model explained this discrepancy [20].

$$\frac{t}{q_t} = \frac{t}{q_e} + \frac{1}{K_2} \cdot \frac{1}{q_e^2} \quad (8)$$

where K_2 is the second order rate constant ($\text{g mg}^{-1} \text{min}^{-1}$).

The straight-line plots of $\log(q_e - q_t)$ versus t for the adsorption of MB onto FB and LS are tested to obtain the rate parameters. The K_1 , $(q_e)_{\text{cal}}$ and correlation coefficients R^2 under different MB concentrations are calculated from these plots and are listed in Table 2. The correlation coefficients R^2 listed in Table 2. for the first-order kinetic model are $R^2 \leq 0.378$ for adsorption of MB onto FB and $R^2 \leq 0.301$ for adsorption onto LS. Since these values of R^2 are low, pseudo first-order plots are not shown as figures.

Table 2. Kinetic Modelling of MB adsorption onto FB and LS at 25°C

	Conc (mg/L)	First Order			Second Order		
		K_1 (min ⁻¹)	q_e (mg/g)	R^2	K_2 (g mg ⁻¹ min ⁻¹)	q_e (mg/g)	R^2
French green beans	10	0.006	0.394	0.050	0.21	1.89	0.998
	20	0.009	1.27	0.2390	0.056	3.4	0.995
	50	0.017	2.96	0.378	0.026	7.825	0.996
	100	0.018	2.3	0.258	0.057	18.87	0.999
Lotus Stem	10	0.002	0.233	0.008	0.45	2.56	0.999
	20	0.007	0.755	0.102	0.115	3.42	0.997
	50	0.011	1.04	0.105	0.091	9.17	0.999
	100	0.016	2.4	0.301	0.066	18.5	0.999

Fig. 7-8 show plots between (t/q_t) versus t . The K_2 and $(q_e)_{cal}$ values, computed from Eq. (13) along with correlation coefficients R^2 are listed in table 2. The R^2 values obtained for this model are above 0.996 for adsorption of MB onto FB and are above 0.9997 for adsorption of MB onto LS for all concentrations studied, which were higher than the R^2 values obtained for the pseudo-first order model. These results imply that the adsorption systems studied follow the pseudo-second order kinetic model at all time intervals.

4. Conclusion

The results of present investigation show that French green beans and Lotus stem low cost materials are suitable for the removal of Methylene Blue from its aqueous solution.. The maximum monolayer adsorption capacity calculated from the Langmuir isotherm model for Lotus stem was found to be 62.5 mg/g while the model is inadequate to determine the adsorption capacity in case of French green beans. Pseudo-second order kinetic model was followed by both the Lotus Stem and French green beans. These results indicate that French Green Beans and Lotus Stem can be used by the Industries discharging Methylene Blue in their effluent, as a cheaper substitute to the more commonly used Activated Carbon. However further experiments need to be conducted to test the dynamic adsorption of Methylene Blue in fixed bed onto French green beans and Lotus Stem.

References

[1] B. Hameed and A. Ahmad, “Batch adsorption of methylene blue from aqueous solution by garlic

peel, an agricultural waste biomass,” *Journal of Hazardous Materials* , pp. 870-875, 2009

[2] H. Deng et al., “Adsorption of methylene blue on adsorbent materials produced from cotton stalk,” *Chemical Engineering Journal* , pp. 326-334, 2011

[3] B. Yasemin and A. Haluk, “A kinetics and thermodynamics study of methylene blue adsorption on wheat shells,” *Desalination* , pp. 259-267, 2006.

[4] J. Panswed and S. Wongchaisuwan “Mechanism of dye wastewater color removal by magnesium carbonate-hydrated basic,” *Water Science and Technology* , pp. 139-144, 1986.

[5] M.M. El-Sheekh, M.M. Gharieb and G.W. Abou-El-Souod, “Biodegradation of dyes by some green algae and cyanobacteria,” *International Biodeterioration & Biodegradation* , pp. 699-704, 2006

[6] G. Ciardelli, L. Corsi and M. Marucci, “Membrane separation for wastewater reuse in the textile industry,” *Resources, Conservation and Recycling* , pp. 189-197, 2000

[7] N. Amri et al., “Waste-based Activated Carbon,” in *IEEE Symp. Humanit. Sci. Eng. Res.*, pp. 33–38, 2012

[8] B.H. Hameed, D.K. Mahmoud and A.L. Ahmad, “Sorption equilibrium and kinetics of basic dye from aqueous solution using banana stalk waste,” *Journal of Hazardous Materials*, pp. 499–506, 2008

- [9] S. Maiti, S. Purakayastha, and B. Ghosh, “Production of low-cost carbon adsorbents from agricultural wastes and their impact on dyes and adsorption,” *Chemical Engineering Communications*, pp. 386-403, 2007
- [10] M.I. El-Khaiary et al., “Adsorption of methylene blue from aqueous solution by chemically treated water hyacinth,” *Toxicological & Environmental Chemistry*, pp. 1079-1094, 2009
- [11] J. Song, et al., “Adsorption characteristics of methylene blue by peanut husk in batch and column modes,” *Desalination*, pp. 119-125, 2009
- [12] R. Lakshmi pathy and N.C. Sarada, “Adsorptive removal of basic cationic dyes from aqueous solution by chemically protonated watermelon (*Citrullus lanatus*) rind biomass,” *Desalination and Water Treatment*, pp. 1-10, 2007
- [13] I. Langmuir, “Constitution and fundamental properties of solids and liquids,” *Journal of the American Chemical Society*, pp. 2221-2295, 1916
- [14] T. Weber and P. Chakkravorti, “Pore and solid diffusion models for fixed-bed adsorbers,” *AIChE Journal*, pp. 220-228, 1974
- [15] Freundlich and H.M.F, “Über die adsorption in losungen,” *Zeitschrift für Physikalische Chemie*, 385-470, 1906
- [16] F. Haghseresht and G. Lu, “Adsorption characteristics of phenolic compounds onto coal-reject-derived adsorbents,” *Energy & Fuels*, pp. 1100-1107, 1998
- [17] A. Kausar, H.N. Bhatti and G. MacKinnon, “Equilibrium, kinetic and thermodynamic studies on the removal of U(VI) by low cost agricultural waste,” *Colloids and Surfaces B*, pp. 124-133, 2013
- [18] K. Foo and B. Hameed, “Insights into the modeling of adsorption isotherm systems,” *Chemical Engineering Journal*, pp. 2-10, 2010
- [19] S. Lagergren and K. Sven, “Zur theorie der sogenannten adsorption gelöster stoffe,” *Vetenskapsakad. Handl*, pp. 1-39, 1898
- [20] Y. Ho, D. Wase, C. Forster, “Kinetic studies of competitive heavy metal adsorption by sphagnum moss peat,” *Environmental Technology*, pp. 71-77, 1966